

Cron Jobs

A **cron job** is a scheduled task, typically a command or script that runs automatically at specified times or intervals on Unix or Unix-like systems like Linux or macOS

- Use cron to schedule script execution

Command	Description
<code>crontab -e</code>	Edit your crontab (create or modify tasks)
<code>crontab -l</code>	List your current cron jobs
<code>crontab -r</code>	Remove all your cron jobs

Why use cron jobs?

Cron jobs automated repetitive tasks, ensuring they run at scheduled times without manual intervention.

Standard Cron (Five-Field Format)

A traditional cron job expression includes **five time fields**, followed by the command to execute:

* * * * * command-to-run

| | | | |

| | | | └ Day of Week (0–6; 0 = Sunday)

| | | └ Month (1–12 or JAN–DEC)

| | └ Day of Month (1–31)

| └ Hour (0–23)

└ Minute (0–59)

Additional Resources and Examples

- **Hello World Cron:** A basic example that echoes "Hello World" daily at noon. [TecAdmin](#)

♦ Example in an alarm script

```
#!/bin/bash
export DISPLAY=:0
export DBUS_SESSION_BUS_ADDRESS=unix:path=/run/user/$(id -u)/bus
```

```
notify-send "Alarm" "Time to check your document!"
paplay /usr/share/sounds/freedesktop/stereo/complete.oga
xdg-open /home/user/Documents/reminder.pdf
```

Sound Effect

```
ls /usr/share/sounds/freedesktop/stereo
```

AT Command:

The at command in Linux is used to schedule a one-time task to run at a specific time in the future.

It is different from cron, which is used for repeating tasks.

at is used when you want a command to run only once at a specified time.

What at Command Does:

- Executes a task once at a given time
- Accepts flexible time formats
- Runs commands in the background
- Uses an atd daemon to process scheduled jobs

Ex:

at now + 10 minutes

at now + 1 hour

at midnight

at teatime

at 3pm + 2 days

Ex:

at 17:00

notify-send "Wake up!"

Ctrl + D